Self-reported food intolerance and mucosal reactivity after rectal food protein challenge in patients with rheumatoid arthritis.

OBJECTIVES: A dietary link to rheumatoid arthritis (RA) has been suspected and an influence on arthritic symptoms by different diets has been reported. Our primary aim was to record the self-experienced adverse food reactions in patients with RA. A secondary aim was to relate self-experienced adverse reactions to dairy produce and wheat to the local mucosal reactivity observed after rectal challenge with cow's milk protein (CM) and wheat gluten. METHODS: A questionnaire about self-experienced adverse reaction to food was sent to 347 RA patients. Rectal challenge with CM and gluten was performed in 27 of these patients and in healthy controls (n = 18). After a 15-h challenge the mucosal production of nitric oxide (NO) and the mucosal release of myeloperoxidase (MPO) and eosinophil cationic protein (ECP) were measured by using the mucosal patch technique. RESULTS: Twenty-seven per cent of the RA patients reported food intolerance (FI) to various foods, and in particular to CM, meat, and wheat gluten. Strong mucosal reactivity to CM was observed in 11% of the patients. Moderately increased mucosal reactivity to CM and gluten was found in 22% and 33%, respectively, of the patients. No relationship was found between self-experienced adverse reactions to CM or gluten and mucosal reactivity to these proteins. CONCLUSIONS: Perceived FI is reported frequently by RA patients, with a prevalence similar to that reported previously in the general population. Mucosal reactivity to CM and gluten is seen in a minor fraction of RA patients and is not related to the frequently perceived intolerance to these proteins.